



ENGINEERING MATHEMATICS-II

Department of Science and Humanities

ENGINEERING MATHEMATICS-II

UNIT 1 : INTEGRAL CALCULUS

class 3: Change of order of
integration

CHANGE OF ORDER OF INTEGRATION

If, in the given double integral the integration is w.r.t. x and then w.r.t y , the process of converting the order of integration is called change of order of integration. Change of order of integration changes the limits of integration.

Consider a double integral $\int \int_R f(x, y) dx dy$ where R is region.

Assume that R lies between the lines $x = x_0$, $x = x_1$ and curves $y = f_1(x)$ and $y = f_2(x)$. For points of R , x lies in the interval $[x_0, x_1]$, y varies between $f_1(x)$ and $f_2(x)$, where $f_1(x)$ and $f_2(x)$ are the ordinates of the points at which the boundary of R is intersected by line through (x, y) and parallel to y -axis.

DOUBLE INTEGRAL

$$\int \int_R f(x, y) dx dy = \int_{x=x_0}^{x_1} \left[\int_{y=f_1(x)}^{f_2(x)} f(x, y) dy \right] dx$$

By change of order of integration, limits of y will be constants y_0, y_1 and x varies between $g_1(y)$ and $g_2(y)$ which the boundary is intersected by the line through (x, y) and parallel to x -axis.

$$\int \int_R f(x, y) dx dy = \int_{y=y_0}^{y_1} \left[\int_{x=g_1(y)}^{g_2(y)} f(x, y) dx \right] dy.$$

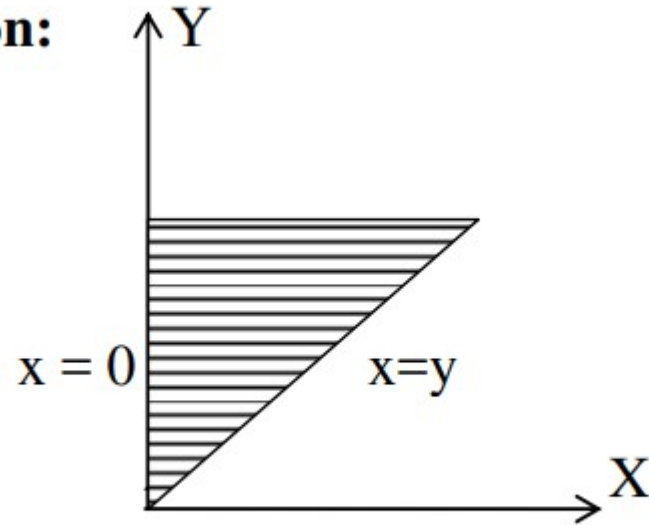
PROCEDURE FOR CHANGE OF ORDER OF INTEGRATION

1. Identify the variables for the limits.
2. Trace the curve.
3. If we are evaluating with respect to y first, then take strip parallel to y -axis.
If the evaluation is with respect to x first, then take strip parallel to x -axis.
4. Rotate the strip to 90° in anti clock wise direction and identify the starting and ending points of the strip, which will be below and upper units of that variable.
5. Identify the limits for other variables for the region of consideration.
6. Evaluate the double integral with new order of integration.

EXAMPLES OF CHANGE OF ORDER OF INTEGRATION

Evaluate $\int_0^\infty \int_0^y y e^{-\frac{y^2}{x}} dx dy$ by changing the order of integration.

Solution:



Given $x=0$, $x = y$, $y = 0$, $y = \infty$.

By changing the order of integration $y: x \text{ to } \infty$, $x : 0 \text{ to } \infty$

2D JACOBIAN

$$\begin{aligned}\int_0^\infty \int_0^y ye^{-\frac{y^2}{x}} dx dy &= \int_0^\infty \int_x^\infty ye^{-\frac{y^2}{x}} dy dx \\ &= \int_0^\infty \int_x^\infty ye^{-\frac{y^2}{x}} d\left(\frac{y^2}{2}\right) dx \\ &= \frac{1}{2} \int_0^\infty \left[\frac{e^{-\frac{y^2}{x}}}{-1/x} \right]_x^\infty dx = \frac{1}{2} \int_0^\infty xe^{-x} dx\end{aligned}$$

Take $u = x, dv = e^{-x} dx$ implies $du = dx, v = -e^{-x}$,

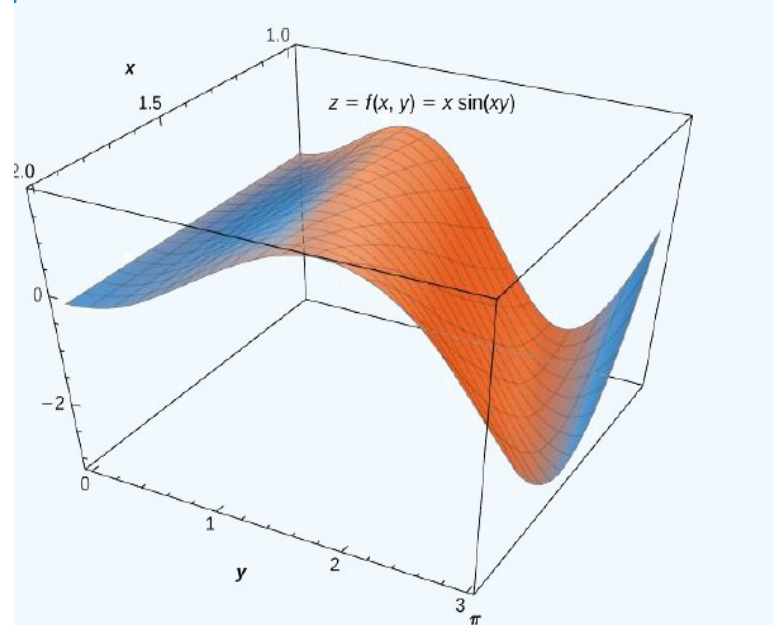
by integration by parts,

$$= \frac{1}{2} \left[x \left(\frac{e^{-x}}{-1} \right) - e^{-x} \right]_0^\infty = \frac{1}{2}$$

FUBINI'S THEOREM FOR RECTANGULAR REGION

Consider the double integral $\iint_R x \sin(xy) dA$ over the region $R = \{(x, y) \mid 0 \leq x \leq \pi, 1 \leq y \leq 2\}$

- Express the double integral in two different ways.
- Analyze whether evaluating the double integral in one way is easier than the other and why.
- Evaluate the integral.



FUBINI'S THEOREM FOR NON RECTANGULAR REGION

a. We can express $\iint_R x \sin(xy) dA$ in the following two ways: first by integrating with respect to y and then with respect to x ; second by integrating with respect to x and then with respect to y .

$$\iint_R x \sin(xy) dA = \int_{x=0}^{x=\pi} \int_{y=1}^{y=2} x \sin(xy) dy dx$$

Integrate first with respect to y .

$$= \int_{y=1}^{y=2} \int_{x=0}^{x=\pi} x \sin(xy) dx dy$$

Integrate first with respect to x .

FUBINI'S THEOREM FOR NON RECTANGULAR REGIONS

b. If we want to integrate with respect to y first and then integrate with respect to x , we see that we can use the substitution $u = xy$, which gives $du = x dy$. Hence the inner integral is simply $\int \sin u du$ and we can change the limits to be functions of x ,

$$\iint_R x \sin(xy) dA = \int_{x=0}^{x=\pi} \int_{y=1}^{y=2} x \sin(xy) dy dx = \int_{x=0}^{x=\pi} \left[\int_{u=x}^{u=2x} \sin(u) du \right] dx.$$

However, integrating with respect to x first and then integrating with respect to y requires integration by parts for the inner integral, with $u = x$ and $dv = \sin(xy)dx$

Then $du = dx$ and $v = -\frac{\cos(xy)}{y}$, so

$$\iint_R x \sin(xy) dA = \int_{y=1}^{y=2} \int_{x=0}^{x=\pi} x \sin(xy) dx dy = \int_{y=1}^{y=2} \left[-\frac{x \cos(xy)}{y} \Big|_{x=0}^{x=\pi} + \frac{1}{y} \int_{x=0}^{x=\pi} \cos(xy) dx \right] dy.$$

Since the evaluation is getting complicated, we will only do the computation that is easier to do, which is clearly the first method.

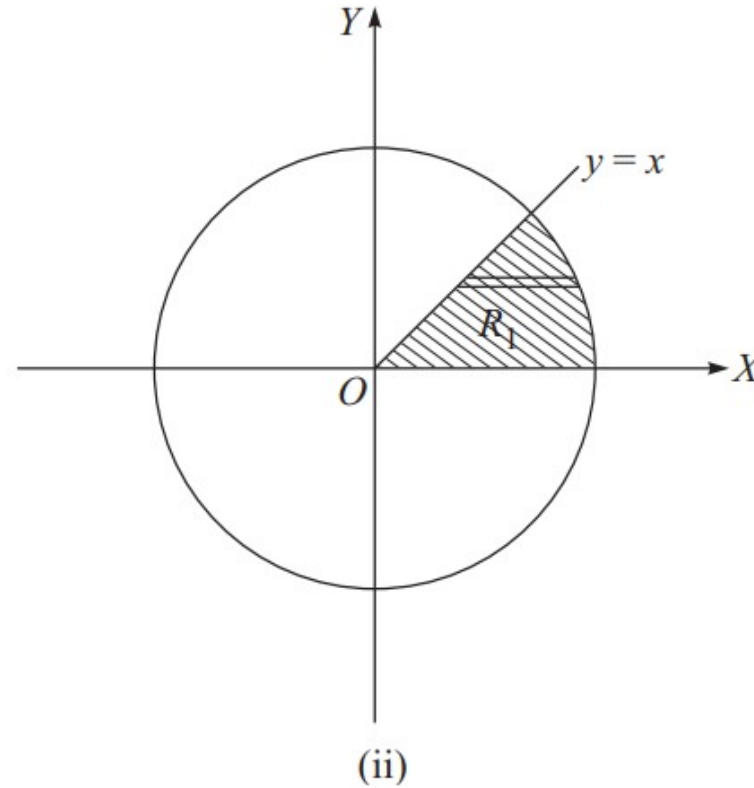
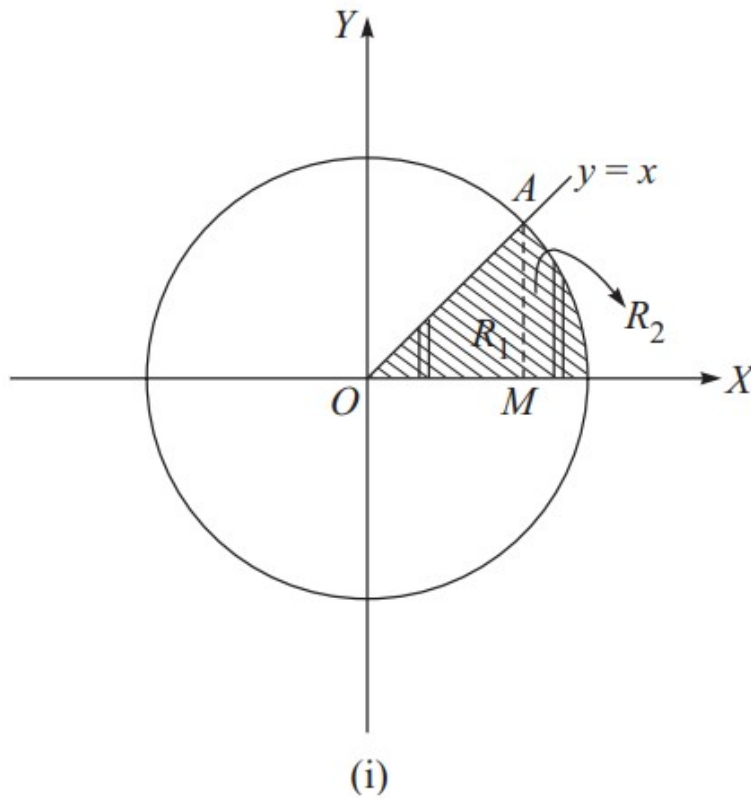
APPLICATIONS OF DOUBLE INTEGRALS

c. Evaluate the double integral using the easier way.

$$\begin{aligned}\iint_R x \sin(xy) dA &= \int_{x=0}^{x=\pi} \int_{y=1}^{y=2} x \sin(xy) dy dx \\&= \int_{x=0}^{x=\pi} \left[\int_{u=x}^{u=2x} \sin(u) du \right] dx = \int_{x=0}^{x=\pi} \left[-\cos u \right]_{u=x}^{u=2x} dx \\&= \int_{x=0}^{x=\pi} (-\cos 2x + \cos x) dx \\&= \left(-\frac{1}{2} \sin 2x + \sin x \right) \Big|_{x=0}^{x=\pi} = 0.\end{aligned}$$

PROPERTIES OF DOUBLE INTEGRALS

Express as a single integral $\int_0^{a/\sqrt{2}} \int_0^x x \, dy \, dx + \int_{a/\sqrt{2}}^a \int_0^{\sqrt{a^2-x^2}} x \, dy \, dx$ and evaluate it.



PROPERTIES OF DOUBLE INTEGRALS

Solution: Let $I_1 = \int_0^{a/\sqrt{2}} \int_0^x x \, dy \, dx$ and $I_2 = \int_{a/\sqrt{2}}^a \int_0^{\sqrt{a^2-x^2}} x \, dy \, dx$

Let R_1 and R_2 be the regions over which I_1 and I_2 are being integrated, respectively and are shown by the shaded region in Fig. 4.4(i).

Also from Fig. 4.4(ii), it is clear that

$$R = R_1 + R_2$$

$$\therefore I = I_1 + I_2 = \iint_R x \, dx \, dy$$

For evaluating I , change the order of integration and then take an elementary strip parallel to the x -axis from

$y = x$ to $y = \sqrt{a^2 - x^2}$, i.e., the circle $x^2 + y^2 = a^2$. Thus,

$$\begin{aligned} I &= \int_0^{a/\sqrt{2}} \int_y^{\sqrt{a^2-y^2}} x \, dx \, dy = \int_0^{a/\sqrt{2}} \left[\int_y^{\sqrt{a^2-y^2}} x \, dx \right] dy \\ &= \int_0^{a/\sqrt{2}} \frac{x^2}{2} \Big|_y^{\sqrt{a^2-y^2}} dy = \frac{1}{2} \int_0^{a/\sqrt{2}} (a^2 - y^2 - y^2) dy = \frac{1}{2} \left[a^2 y - \frac{2y^3}{3} \right]_0^{a/\sqrt{2}} = \frac{a^3}{3\sqrt{2}} \end{aligned}$$



THANK YOU
